

SUCCESS CRITERIA

Success Criteria (Category)	Details	Criterion (Success Measure)	Justification & Research
<u>Aesthetics</u> i. The website should be visually appealing, neat, and organized. There must be a visual representation of the website. ii. The website should mostly have the colors to match the colors of steel. This will give a modern look.	i. The website should consist of tabs for organization, and the fonts of the website should also be uniformed. There should be specific sections of the website to make the website organized. ii. It is important that my website is visually appealing and organized so users feel comfortable and engaged. I will use mono-tone colors (white, black, and grey) to match the steel theme and create a modern look.	i. ii. Through a survey, at least 20 individuals should agree that the website is organized and visually appealing (composed of mono-tone colors such as white, black, and grey). - Do you think the website is organized? If yes, how is this website organized? - What are the major colors that you see on this website?	According to Hiilite, a website is visually appealing when it has a substantial layout and displays with clear graphics and photos. Especially the fonts should be a visual point for the audience at first sight. The author claimed that colors and fonts should complement each other with the combination of real life images. The author also mentions the importance of a structured theme as it should describe the mood of the website. In a website, there are mainly three different layouts: fixed, liquid, and mobile layout. A fixed layout is a layout often organized in columns either horizontally or vertically across the screen. On the other hand, a liquid layout is mostly for reading and with articles. 1. Always come up with an outline that has the purpose and the functionality of the site and pick the necessary choices of design (Hiilite). “Creating Aesthetically Pleasing Websites: 5 Elements To Consider.” Hiilite, 2 May 2022, https://hiilite.com/creating-aesthetically-pleasing-websites/. Accessed 10 May 2025.
<u>Function</u> i. The website should serve the purpose of calculations of steel optimization and maximization. ii. The website should introduce the steel industry to an unfamiliar audience. Iii. The website should explain the reasoning and the objective of steel optimization.	i. The whole purpose of this website is to provide optimization for the steel industry, so the website should have a clear purpose. ii. I want the website to be open to some unfamiliar audience which means that there should be a detailed introduction page of what steel is and what steel optimization is. Furthermore, it should share the objective of this steel optimization and tie this to real life.	I will interview a professional my dad to assess the purpose, objective, and the introduction of steel based on the success criteria. Since my dad is in the steel industry, he will assess the composition of this website.	Writing content inside a website: 1. The contents should be written as an easy-to-read format ⇒ like a conversation in order to engage the audience into the website 2. You will always have to keep the audience and the purpose in mind. 3. Be short and direct ⇒ this will give emphasis ⇒ a paragraph should be 70 words or less 4. The audience should feel the content as personal ⇒ avoid bureaucratic language which is mostly tone down. 5. Users only read 20% of the content, so it should be kept short and concise. a. This is why subheadings and titles are important ⇒ it has to sum up the whole content. https://www.umaryland.edu/cpa/website-manual/prepare/web-writing/
<u>Program</u> i. The optimization	i. While optimizing the maximum limit of the	i. To evaluate the accuracy and the effectiveness of the steel optimization program, I will try at least 5 trial cases that use both the conventional	Coding with math involves using both mathematical concepts and expressions together.

algorithm should be more effective and customizable than other cutting methods conventionally. ii. The user should be able to cut/add dimensions and constraints that are needed for steel cutting. iii. These results should be saved that could be exported into another device.	steel cutting process, the program should be 97% accurate to the input the workers are putting in the values for. ii. This will be a customization concept for the users to adjust the length of the steel freely. iii. In order to make the website user friendly, the program will have to include and save results from the previous input/output.	way and the program that I have created. The website will be considered “successful” if the algorithm achieves at least 97% of steel. ii. Users will also test the input system of custom lengths and constraints. At least 5 users should be able to add or adjust dimensions. iii. The saved results should be exported to PDF format as well.	- In order to perform calculations and real life situations (this is connected to my project) EXAMPLES: $a = 5$ $b = 3$ $\text{sum} = a + b$ # sum will be 8 $\text{difference} = a - b$ # difference will be 2 $\text{product} = a * b$ # product will be 15 $\text{quotient} = a / b$ # quotient will be 1.666... $\text{power} = a ** b$ # power will be 125 These are simple algebraic concepts that may be used in real life. In order to understand this, the programmer will have to understand the real life situation thoroughly. 1. Plan out a scenario 2. Work out a logic 3. Fit the logic into various scenarios (mostly if...statements) Sometimes, we may incorporate algorithms as well if it's necessary for the real life situation. https://www.thecodingspace.com/blog/2024-01-06-is-coding-math-exploring-the-relationship-between-programming-and-mathematics/
<u>User Experience</u> i. The navigation between different sections should be efficient for the users to shift tabs. ii. There should be a guide for the users who are using this website for the first time.	i. On the top of the website should be a whole insert of indicators that indicates the users with the certain pages such as “How to Page”, “Calculator” “Why” pages ii. The guide should be on the How To Page on a separate page of tabs in order to enhance the user’s experience through clarity.	I. A survey will be sent and at least 10 first time individuals will have to achieve more than 90% answers that they can easily navigate through tabs such as “How To,” “Calculator,” and “Why.” - Were you able to find the specific features easily? - How clear was the How-To page to help your understanding of the website?	According to research, users want a website that is fast, user-friendly, reliable, and provides clear, relevant information. It is also an important factor that a website is a mobile-friendly website and offers personal experience. Since users do not want to spend a long period of time on the website, efficiency is key as well. https://1827marketing.com/smart-thinking/9-principles-good-web-design What is an example of a user-friendly website? - One of the most prominent examples is Airbnb. This website is known for its convenient use of navigation and responsive designs. Users can easily find accommodations which is the purpose of Airbnb. - From this, rather than the website going off task, I will make the website all correlated to the purpose. What makes a Website User-Friendly? 1. Simplifying the website’s structure 2. All content should be accessible 3. Loading speed $\Rightarrow$ convenience 4. Mobile responsiveness https://www.feedthebot.org/blog/user-friendly-website/#:~:text=Examples%20of%20User%2DFriendly%20Websites%20*%20Airbnb,

			%20For for%20its%20accessible%20and%20organized%20content%20structure.
<u>Business Value:</u> i. The website should demonstrate the estimated measure of cost savings that updates. ii. The website should ensure that the environmental impact is quantified. iii. The website should be accessible for individuals	i. Each month, the website should display on the first welcome page to indicate the impact this optimization has given to the steel industry. ii. In one of the “why” pages, there should be the reason behind why steel optimization is necessary. It should state the cost reduction and the CO₂ reduction that occurred after using this optimization system. iii. It is important that the steel optimization program is adapted and accessible to the companies in the steel industry.	The homepage will include a section that shows monthly updates on the saved steel, reduction of costs, and the estimated CO₂ emission. The success will look like 80% of users indicating that they understood the real-world value of optimization after reading the Why page. - Did the page help you understand the benefits of optimization? - Was the environmental impact clearly shown or explained? iii. At least one company should adapt this website while using the steel optimization program. This looks like cost reduction and reducing the amount of CO₂ emission companies are burning. This means that the feedback from the industry should be positive	In order to communicate the business values on a website, the content should be focused on stating how the code or service BENEFITS the customer. - This should be correlated with the current problem and skeptical aspects that they are experiencing. This will help them gain a better understanding of the model. 1. Credibility a. The legitimacy as a website is crucial ⇒ making a great first impression 2. Brand a. Showcase the brand to the customer Therefore, showing measurable outcomes and clear reasons for optimization makes the website relevant and persuasive for a business target. https://www.forbes.com/councils/thevec/2020/02/03/why-every-business-needs-a-website/